Linear regression and causality

ECON726

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October 6, 2026

Housekeeping

No class next week!

Homework

- Project plan due!
- Read MHE 3.3 (Matching + propensity score)
- Optional: chapter 5, section 5.2-5.5 in Causal ML book
- Optional: chapter 5 in Causal Inference: the Mixtape

Today:

- Reviewing regression under unconfoundedness:
  - MHE 3.2.
  - Chapter 2 & 5.1 in Causal ML book

Empirical project

Paper selection: Due **September 28** [5% of final grade]

- Done!

Project plan: Due **October 12** [15% of final grade]

- This paper should be your first effort at addressing points (1) and (2) above. Summarize the public policy or business context of the paper. Write down the specific question the paper is evaluating. Discuss the data your are using (both your replication data, and data for your extension). Verify that you can access the data and replication package. Discuss the extensions you are considering, and, if necessary, where you will be getting that data. Verify that you've read the paper.

Replication results: Due **November 2**: [10% of final grade]

- You should submit some of your main replication results. The final results should be formatted as tables and charts in a word document or PDF, submitted on Brightspace. Your code and data should accessible on GitHub.

Presentation: Due **December 1** & **December 8** [10% of final grade]

- Prepare a 10 minute presentation on your paper. Focus on the research question, main result, and one extension.

Final Paper: Due **December 11** [40% of final grade]

- Your paper should contain 6-10 pages of text, and 3-6 pages of figures (double spaced, 12 point font). You must provide the code and data to replicate your findings. Your final project should be submitted on Brightspace, and your code and data should be uploaded to github (note: if your data is too large to upload to github, let me know).

Potential Outcomes

Conditional Independence Assumption

Regression and causality

Regression gives the best linear approximation to the CEF, in a MSE sense

- But when does a regression represent something causal?

A regression is **causal** when the CEF it approximates is causal

- Helpful to think of a CEF being causal by using the potential outcomes framework
- CEF is causal when it describes differences in average potential outcomes for a fixed reference population

Review - Potential Outcomes framework

Consider the relationship between future earnings Y_i and whether i attends college, denoted by D_i

$D_i \in \{0, 1\}$	Binary **treatment variable** for individual i
Y_i	**Outcome** Measure of earnings for individual i
$Y_i(1)$	**Treatment potential outcome** Earnings if individual i goes to college ($D_i = 1$) Alternative notation: Y_{1i}, Y_i^1, y_{i1}
$Y_i(0)$	**Control potential outcome** Earnings if individual i does not go to college ($D_i = 0$) Alternative notation: Y_{0i}, Y_i^0, y_{i0}

- $Y_i(0)$ is earnings of an individual had they not gone to college, *irrespective of whether they actually went*
- Causal effect of college attendance for individual i is the difference $Y_i(1) - Y_i(0)$

Observed outcome Y_i can be written as:

$$\begin{aligned} Y_i &= \begin{cases} Y_i(1) & \text{if } D_i = 1 \\ Y_i(0) & \text{if } D_i = 0 \end{cases} \\ &= Y_i(0) + (Y_i(1) - Y_i(0))D_i \end{aligned}$$

We can see $Y_i(0)$ or $Y_i(1)$ in the data, but never both

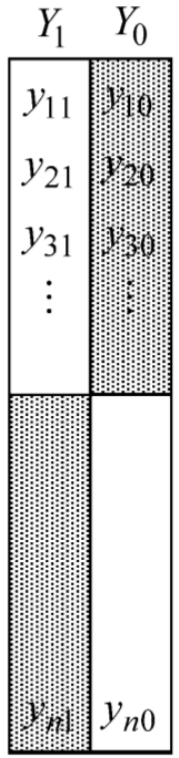
Review - Fundamental problem of causal inference

Problem:

- If an individual is treated ($D_i = 1$), we never observe the outcome in which they are not treated ($Y_i(0)$)
- If an individual is not treated ($D_i = 0$), we never observe the outcome in which they are treated ($Y_i(1)$)

The causal effect $Y(1) - Y(0)$ is unidentifiable

Group	$Y(1)$	$Y(0)$
$D = 1$	Observable as Y	Unobservable counterfactual
$D = 0$	Unobservable counterfactual	Observable as Y



Stable Unit Treatment Value Assumption (SUTVA)

To link potential outcomes to observed data, we need two assumptions; together, these are known as **SUTVA**

1. Consistency: we observe $Y_i = Y_i(D_i)$

- Equivalently, $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$
- Rules out hidden versions of treatment: $D_i = 1$ must mean the same thing for everyone
- e.g., if "college" includes both a 2-year and 4-year degree, $Y_i(1)$ is not well-defined

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2. No interference: i's potential outcomes depend only on i's own treatment, not on anyone else's

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When might no interference fail?

- Vaccines: if many people are vaccinated, herd immunity protects the unvaccinated too
- College: if many people get a degree, the college wage premium may fall

Source: Chernozhukov et al. (2025), Assumptions 2.1.1–2.1.2 and Section 2.4

Types of comparisons

Causal effect of interest is often **Average Treatment Effect**:

$$\mathbf{E}[Y(1) - Y(0)] = \mathbf{E}[Y(1)] - \mathbf{E}[Y(0)]$$

or **Average Treatment Effect on the Treated**:

$$\mathbf{E}[Y(1) - Y(0)|D = 1] = \mathbf{E}[Y(1)|D = 1] - \mathbf{E}[Y(0)|D = 1]$$

We often discuss what is directly observable in data as the **naïve comparison**. In the population this is:

$$\mathbf{E}[Y|D = 1] - \mathbf{E}[Y|D = 0]$$

This is also the **average predictive effect** of treatment status on outcome.

Naïve comparison: $\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0]$

Let's consider the averages of $Y_i(1)$, $Y_i(0)$, and Y_i:

$$\begin{aligned}
\mathbf{E}[Y|D=1] &= \mathbf{E}[Y(0) + (Y(1) - Y(0))D|D=1] \quad \because Y = Y(0) + (Y(1) - Y(0))D \\
&= \mathbf{E}[Y(1)|D=1] \\
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\end{aligned}$$

We can re-write the average predictive effect (observed difference) as:

$$\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0]$$

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&= \mathbf{E}[Y(1) - Y(0)|D=1] + \mathbf{E}[Y(0)|D=1] - \mathbf{E}[Y(0)|D=0]
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=& \underbrace{\mathbf{E}[Y(1) - Y(0)|D=1]}_{\text{Avg. treatment effect on treated}} + \underbrace{\mathbf{E}[Y(0)|D=1] - \mathbf{E}[Y(0)|D=0]}_{\text{Selection bias}}
\end{aligned}$$

Naïve comparison (cont.)

Observed difference in earnings equals the average treatment effect (ATE) on the treated plus selection bias:

$$\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0] = \underbrace{\mathbf{E}[Y(1) - Y(0)|D=1]}_{\text{ATE on treated}} + \underbrace{\mathbf{E}[Y(0)|D=1] - \mathbf{E}[Y(0)|D=0]}_{\text{Selection bias}}$$

ATE on treated is the average causal effect of college attendance on those who went to college:

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Selection bias:

$$\mathbf{E}[Y(0)|D=1] - \mathbf{E}[Y(0)|D=0]$$

What do you think selection bias looks like in this example?

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What do you think selection bias looks like in this example?

- The difference in $Y_i(0)$ between those who went to college and those who didn't
- Consider someone who went to college. Now assume that person didn't go. Their expected earnings may be higher than those of someone who didn't go to college at all

  $\Longrightarrow$ Those who didn't go to college, on average, have lower values of $Y_i(0)$

  $\Longrightarrow$ Selection bias is positive

Example: Marijuana and longevity

Does smoking marijuana (D_i) affect longevity (Y_i)?

- Suppose marijuana has **no causal effect** on longevity: $Y_i(1) = Y_i(0)$ for everyone

$\Longrightarrow$ ATE on treated $= \mathbf{E}[Y(1) - Y(0)|D = 1] = 0$

But smoking is not randomly assigned: people choose whether to smoke

- Suppose smokers are more likely to make other poor health choices (e.g., drinking alcohol) that shorten life expectancy

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Then smokers would have shorter lives *even if they had not smoked*:

$$\underbrace{\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0]}_{\text{Naïve comparison} < 0} = \underbrace{0}_{\text{ATE on treated}} + \underbrace{\mathbf{E}[Y(0)|D=1] - \mathbf{E}[Y(0)|D=0]}_{\text{Selection bias} < 0}$$

- Data show a negative relationship between smoking and longevity, even though the causal effect is zero
- Selection bias is negative here; in the college example it was positive

Source: Chernozhukov et al. (2025), Example 2.1.2

Random assignment and selection bias

Random assignment assumption: treatment status is statistically independent of each potential outcome, and assignment is nontrivial:

$$D \perp Y(d) \text{ for } d \in \{0,1\}, \quad \text{and} \quad 0 < \mathbf{P}(D=1) < 1$$

- Treatment assignment mechanism is purely random; knowing D tells us nothing about $Y(0)$ or $Y(1)$
- **Positivity** ($0 < \mathbf{P}(D=1) < 1$) ensures there are units in both treatment and control

Random assignment ($Y(0), Y(1) \perp D$) will solve the selection problem

- Causes selection bias to equal zero

$$\begin{aligned} & \mathbf{E}[Y(1)|D=1] = \mathbf{E}[Y(1)|D=0] = \mathbf{E}[Y(1)] \\ \text{and} \quad & \mathbf{E}[Y(0)|D=0] = \mathbf{E}[Y(0)|D=1] = \mathbf{E}[Y(0)] \\ & \implies \mathbf{E}[Y(0)|D=1] - \mathbf{E}[Y(0)|D=0] = 0 \\ & \implies \text{Selection bias is } 0 \\ & \implies \text{Naïve comparison = ATE on treated} \end{aligned}$$

The ideal experiment

A useful first question for any causal claim: **what experiment would you run if you could?**

- Who is randomized, into what treatment, and what outcome do you measure?
- If you can't describe the ideal experiment, the causal question may not be well-defined

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Marijuana and longevity:

- Randomly assign a group of people to smoke or not smoke marijuana for decades; compare age at death
- Randomization ensures smokers and non-smokers have the same $\mathbf{E}[Y(0)] \implies$ no selection bias

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College and earnings:

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- Randomly admit (and require attendance from) some high school graduates; compare later earnings

Often the ideal experiment is unethical, infeasible, or too expensive

- Rest of the course: approximating the ideal experiment with observational data

Source: Angrist and Pischke (2009), Chapter 1

Randomization and regression

Assume treatment effects are constant and the same for everyone

$$Y_i = Y_i(0) + (Y_i(1) - Y_i(0))D_i$$

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Evaluate this with the treatment on ($D = 1$) and off ($D = 0$):

$$
\begin{aligned}
D = 1 &\implies \mathbf{E}[Y|D = 1] = \alpha + \beta + \mathbf{E}[\eta|D = 1] \\
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Then:

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Selection bias will equal zero if the correlation between the error term and treatment assignment is zero.

- If D is randomly assigned, regression of Y onto D (and a constant) yields the causal effect β

Covariates and pre-treatment heterogeneity

Suppose in addition to outcome (Y) and treatment (D), there are **pre-treatment** or **pre-determined** covariates W

- e.g. interested in treatment effect across age

Treatment effect and naïve comparison likely will not match if there is pre-treatment heterogeneity

Assume: treatment assignment independent of covariates:

$$D \perp (Y(0), Y(1), W), \quad \text{and } 0 < \mathbf{P}(D = 1) < 1$$

then:

$$\mathbf{E}[Y|D = d, W] = \mathbf{E}[Y(d)|D = d, W] = \mathbf{E}[Y(d)|W]$$

$\implies$ **conditional average treatment effects** can be estimated from conditional average predictive effects:

$$\mathbf{E}[Y(1)|W] - \mathbf{E}[Y(0)|W] = \mathbf{E}[Y|D = 1, W] - \mathbf{E}[Y|D = 0, W]$$

Testing covariate balance

Random assignment assumption induces covariate balance.

- Distribution of covariates should be the same under treatment and control:

$$W|D = 1 \sim W|D = 0$$

equivalently:

$$D|W \sim D$$

Useful implication is:

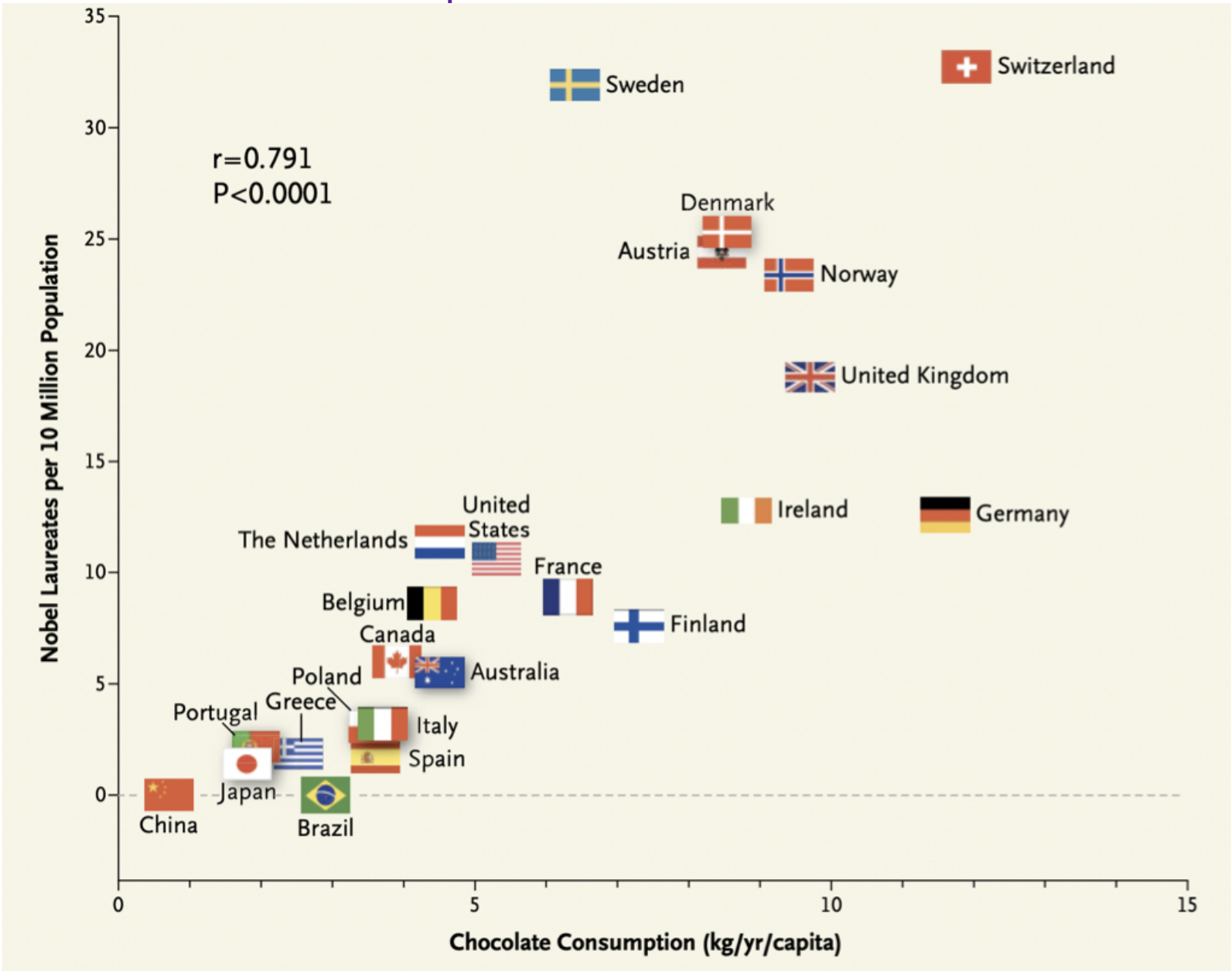
$$\mathbf{E}[D|W] = \mathbf{E}[D]$$

This is testable using regression tools; the R^2 of a regression of D on W is 0 under randomization.

Potential Outcomes

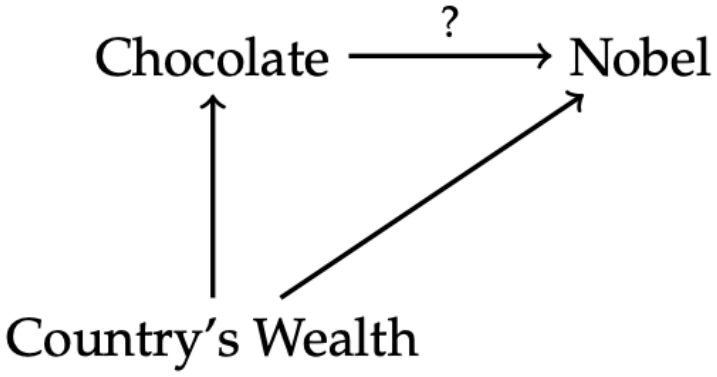
Conditional Independence Assumption

Impact of chocolate consumption on Nobel laureates?



Common cause

Common cause creates non-causal association



- Should be comparing Switzerland to a country similar to Switzerland

Conditional Independence Assumption

Last section: how do we infer causal effects when we have **random assignment** of treatment.

$\Longrightarrow$ Independence of potential outcomes and the treatment

$\Longrightarrow$ $Y(0), Y(1) \perp D$

$\Longrightarrow$ randomization

What happens when we don't have independence? Meaning the occurrence of D influences the occurrence of $Y(0)$ and $Y(1)$ (or vice versa)?

Perhaps we can condition on certain variables X, such that the treatment D_i and the potential outcomes $Y_i(1), Y_i(0)$ LOOK independent

$$\{Y_i(0), Y_i(1)\} \perp D_i | X_i$$

- Think of treatment D_i as generated randomly (independent of potential outcomes) **conditional** on some pre-treatment variable X

$\Longrightarrow$ can learn about average causal treatment effects by regression of Y_i on D_i and X_i.

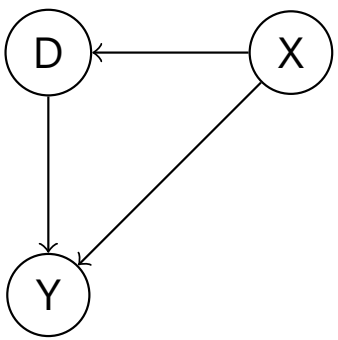
Conditional independence

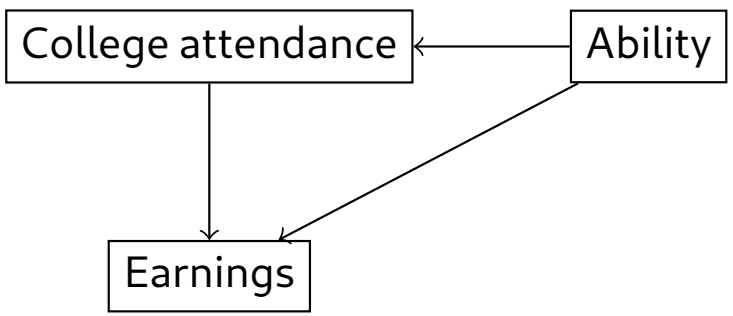
Conditional independence assumption:

$$\{Y_i(0), Y_i(1)\} \perp D_i | X_i$$

Intuitively: occurrence of D_i does not influence the occurrence of $Y_i(0)$ or $Y_i(1)$, or vice versa, conditional on knowing X_i:

$$\mathbf{E}[Y_i(0)|D_i, X_i] = \mathbf{E}[Y_i(0)|X_i]$$
$$\mathbf{E}[Y_i(1)|D_i, X_i] = \mathbf{E}[Y_i(1)|X_i]$$





Selection bias under CIA

Under conditional independence assumption, selection bias disappears conditional on X_i:

$$\begin{aligned}\mathbf{E}[Y_i(0)|X_i, D_i = 1] - \mathbf{E}[Y_i(0)|X_i, D_i = 0] =& \mathbf{E}[Y_i(0)|X_i, D_i = 0] - \mathbf{E}[Y_i(0)|X_i, D_i = 0] \\ =& 0\end{aligned}$$

So we can write the difference in observed outcomes **conditional on** X_i as:

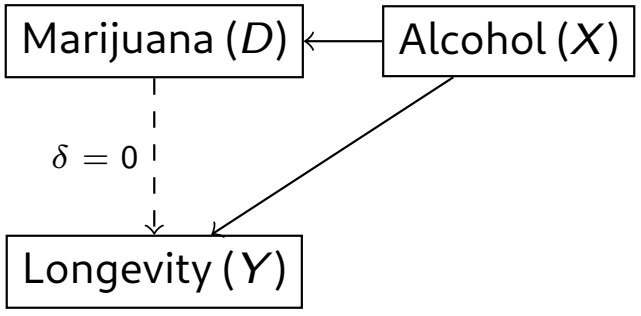
$$\begin{aligned}\mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0] &= \mathbf{E}[Y_i(1)|X_i, D_i = 1] - \mathbf{E}[Y_i(0)|X_i, D_i = 0] \\ &= \mathbf{E}[Y_i(1)|X_i] - \mathbf{E}[Y_i(0)|X_i] \\ &= \mathbf{E}[Y_i(1) - Y_i(0)|X_i]\end{aligned}$$

Given the CIA, comparisons of average earnings (conditional on X_i) have a causal interpretation

Example

Suppose you are studying the impact of marijuana on longevity.

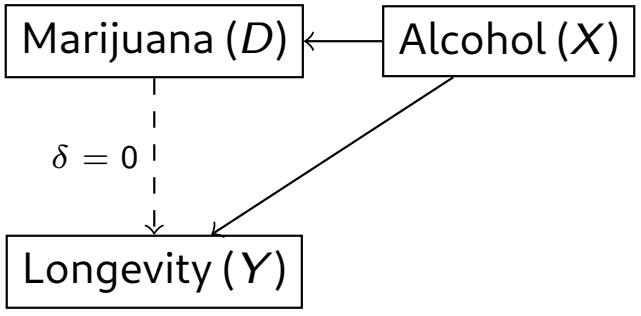
- As before, assume no causal effect:
  $Y = Y(0) = Y(1)$, so
  $\delta = \mathbf{E}[Y(1)] - \mathbf{E}[Y(0)] = 0$
- Smoking decisions (D) are driven by other behaviors X (e.g., drinking alcohol) that shorten life



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Without adjusting for X: higher D predicts higher X, which predicts lower Y

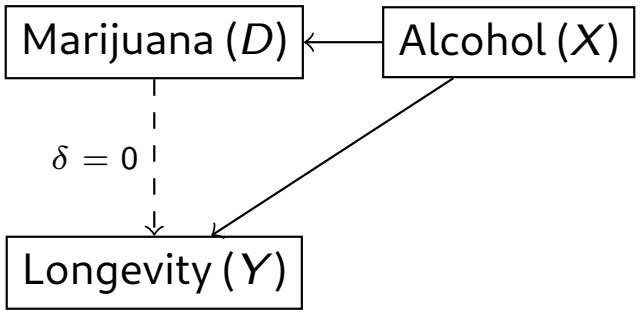
$$\mathbf{E}[Y|D = 1] - \mathbf{E}[Y|D = 0] < 0$$

- Predictive effect $\neq$ causal effect $\implies$ **confounding** (selection bias)

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Conditioning on X removes the selection bias:

$$\mathbf{E}[Y|D = 1, X] - \mathbf{E}[Y|D = 0, X] = \mathbf{E}[Y(1) - Y(0)|X]$$

provided that, conditional on X, variation in D is independent of potential outcomes:

- **Conditional ignorability:** $D \perp Y(d)|X$ for each d (i.e., the CIA)
- **Consistency:** $Y = Y(D)$

Source: Chernozhukov et al. (2025), Section 5.2 and Assumption 5.2.1

Selection on X creates selection bias



Potential outcome $Y(0)$ for everyone. Binary covariate X: units with $X = 1$ ($\times$) tend to have higher outcomes than units with $X = 0$ ($+$)

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

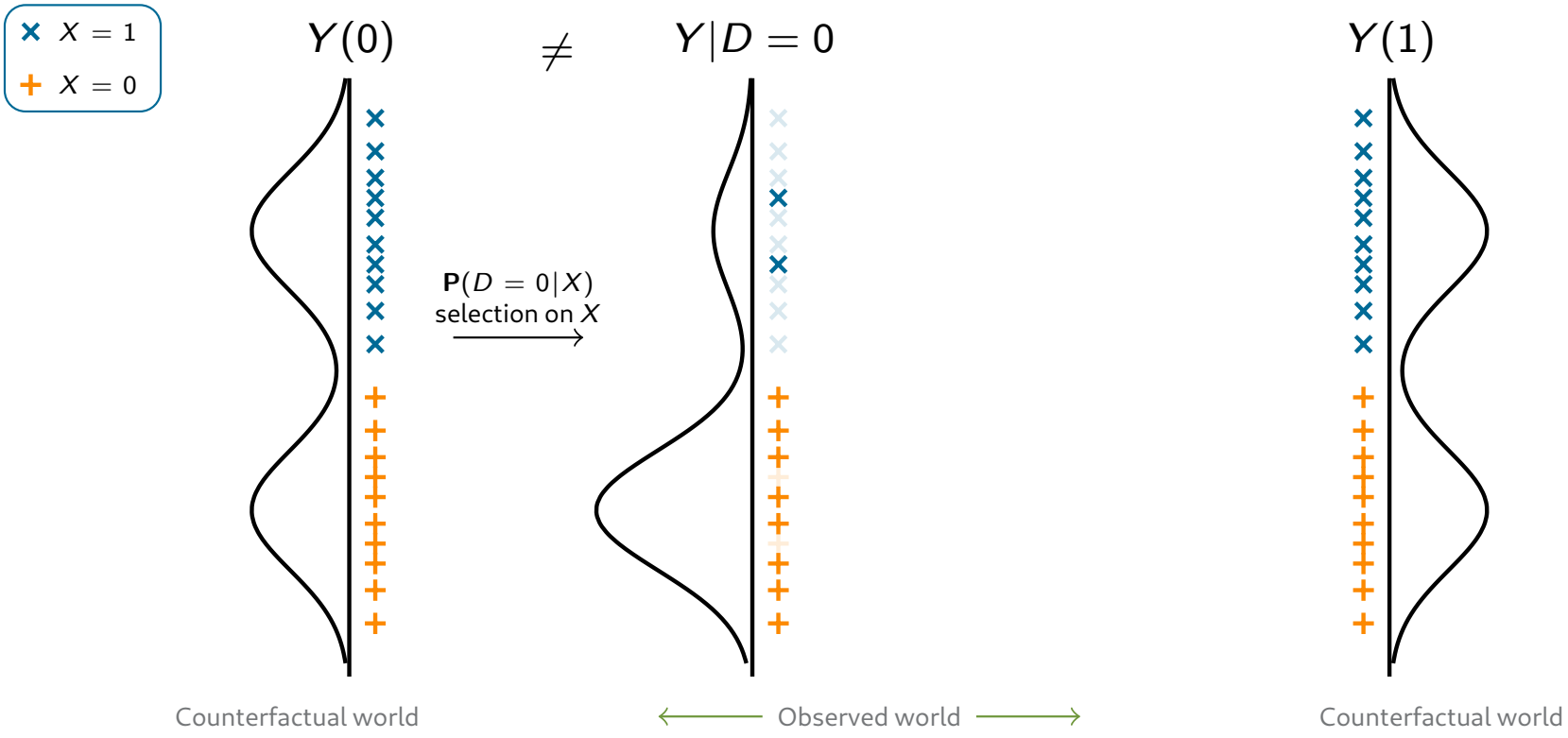
Selection on X creates selection bias



No treatment effect: each unit's $Y(1)$ equals its $Y(0)$, so $Y(0)$ and $Y(1)$ have identical distributions ($\delta = 0$)

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

Selection on X creates selection bias

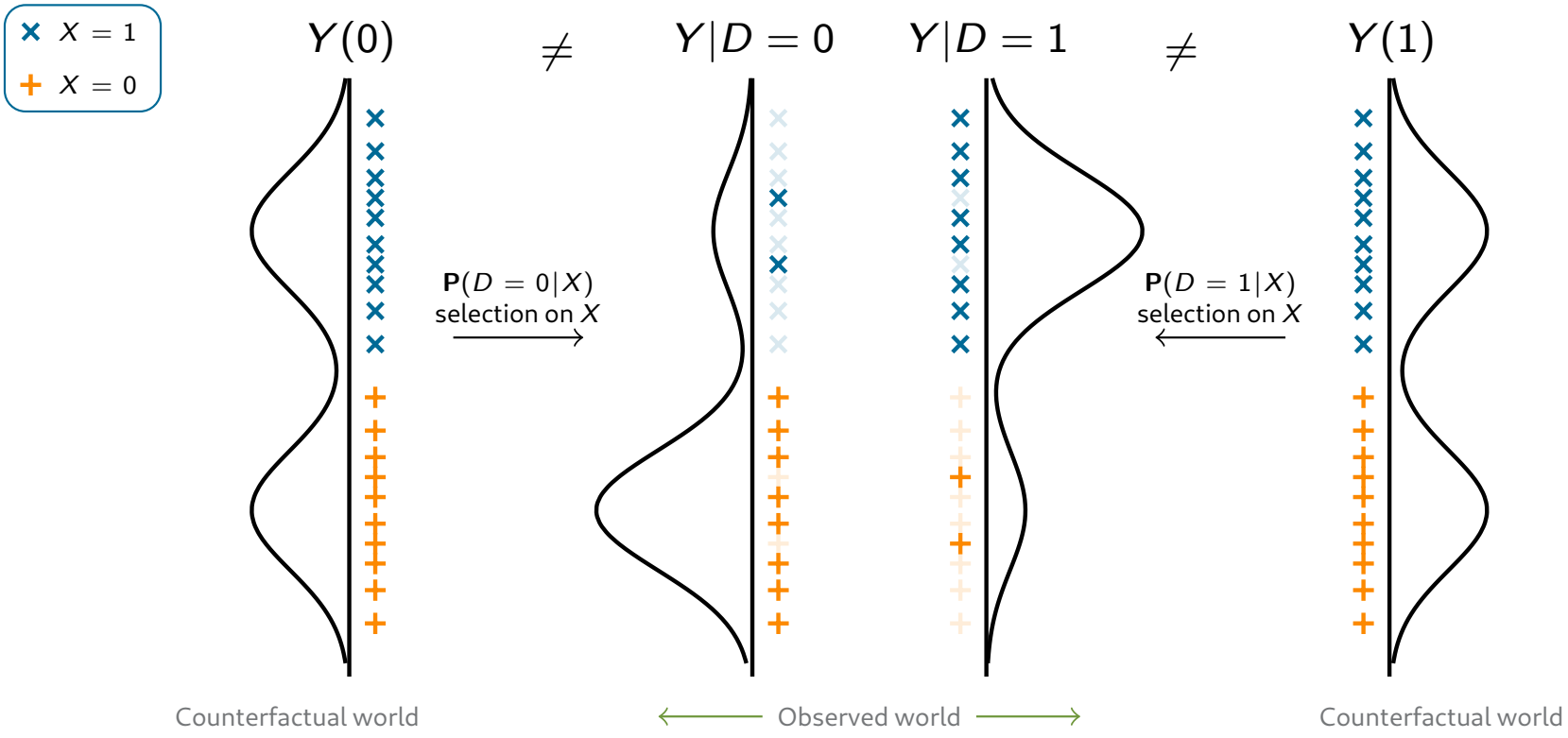


Selection on X: $X = 0$ units are more likely to be untreated, $\mathbf{P}(D = 0|X = 0) > \mathbf{P}(D = 0|X = 1)$

- We only observe $Y(0)$ for the untreated $\implies$ $Y|D = 0$ is shifted *down* relative to $Y(0)$

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

Selection on X creates selection bias

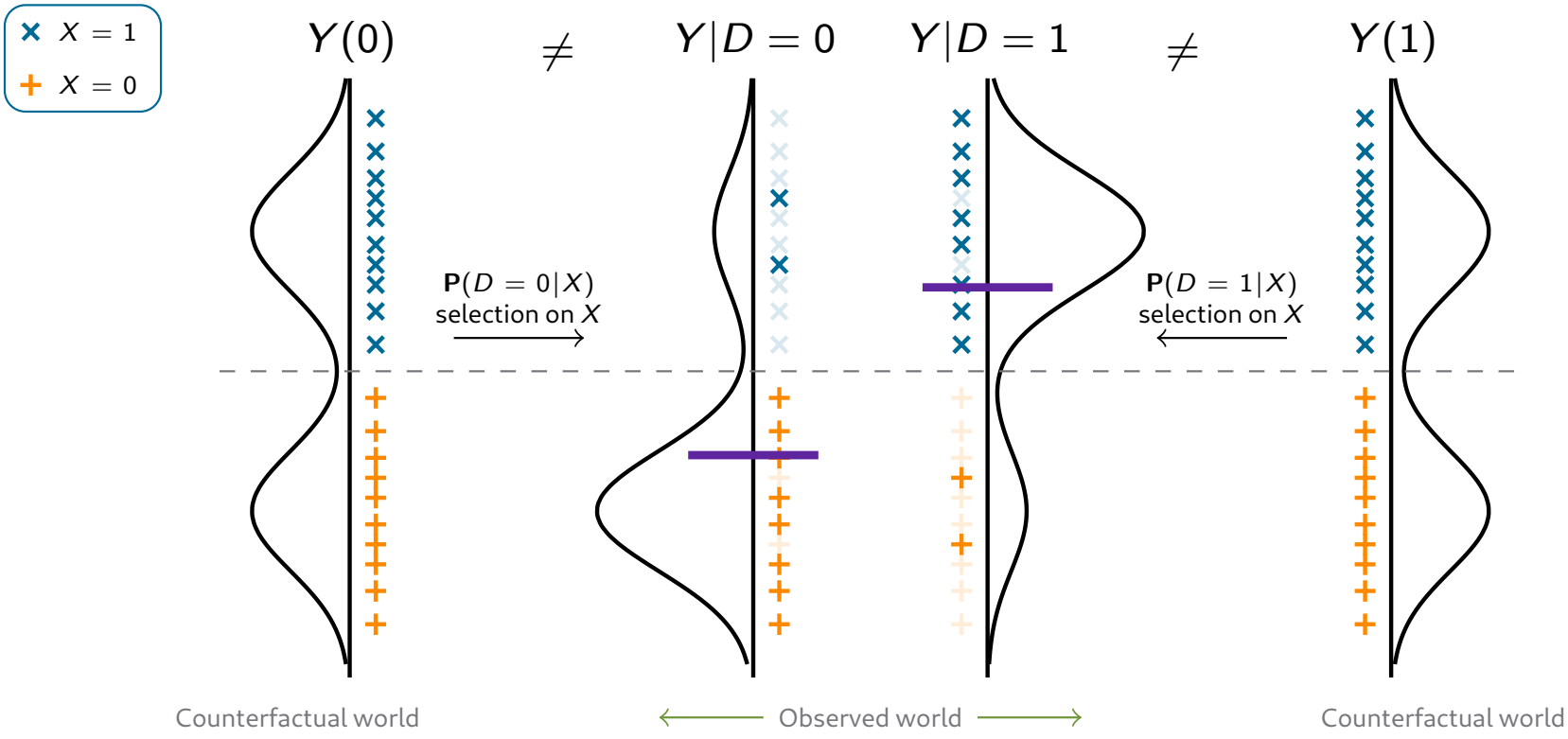


$X = 1$ units are more likely to be treated, $\mathbf{P}(D = 1|X = 1) > \mathbf{P}(D = 1|X = 0)$

- We only observe $Y(1)$ for the treated $\implies$ $Y|D = 1$ is shifted *up* relative to $Y(1)$

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

Selection on X creates selection bias



Naïve comparison: $\mathbf{E}[Y|D = 1] > \mathbf{E}[Y|D = 0]$, even though $\mathbf{E}[Y(1)] = \mathbf{E}[Y(0)]$ (dashed line)

- Difference is entirely **selection bias**: the treated and untreated have different mixes of X

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

Conditioning on X removes selection bias

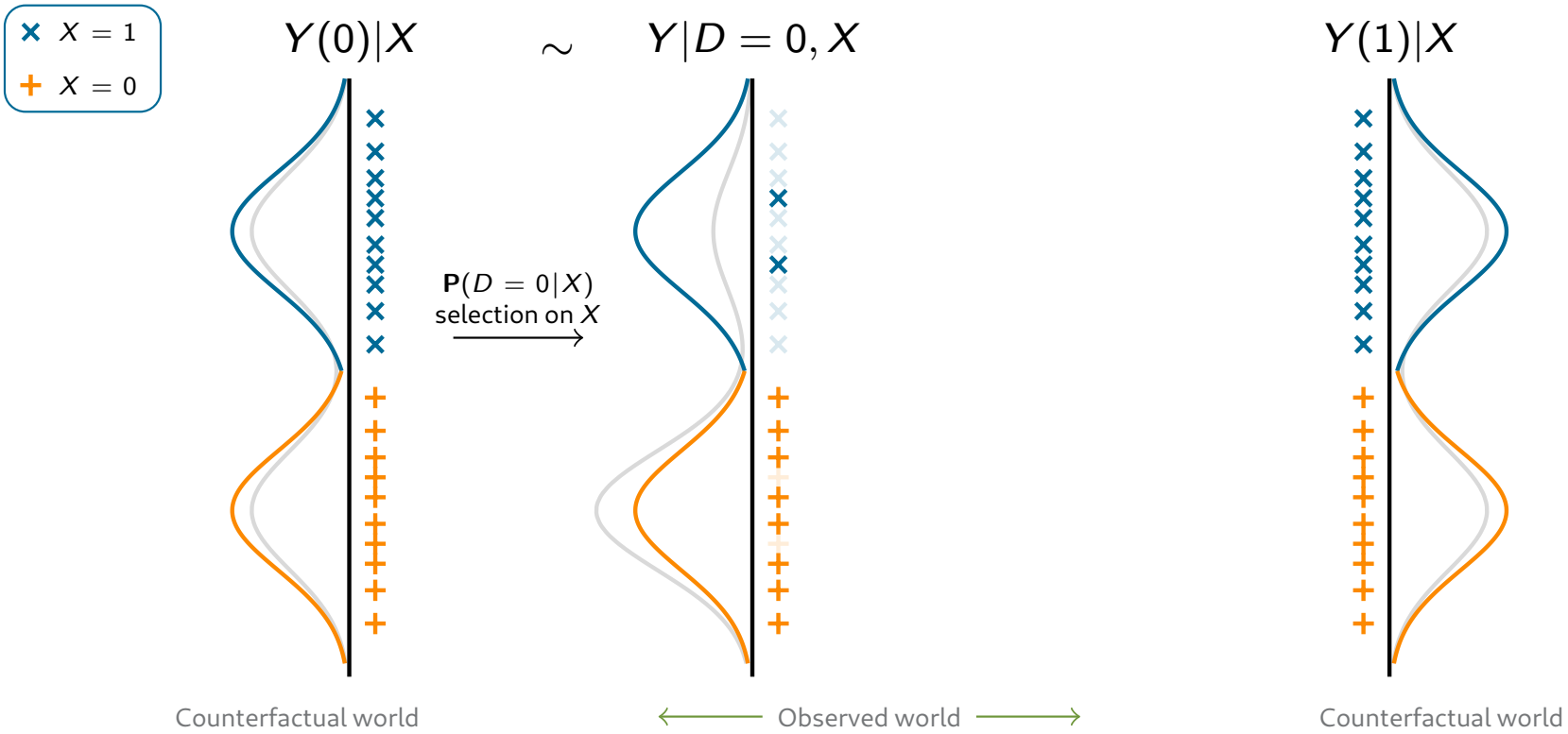


Condition on X: split each potential outcome distribution by X

- $Y(0)|X = 1$ (blue) and $Y(0)|X = 0$ (orange); same for $Y(1)$

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

Conditioning on X removes selection bias

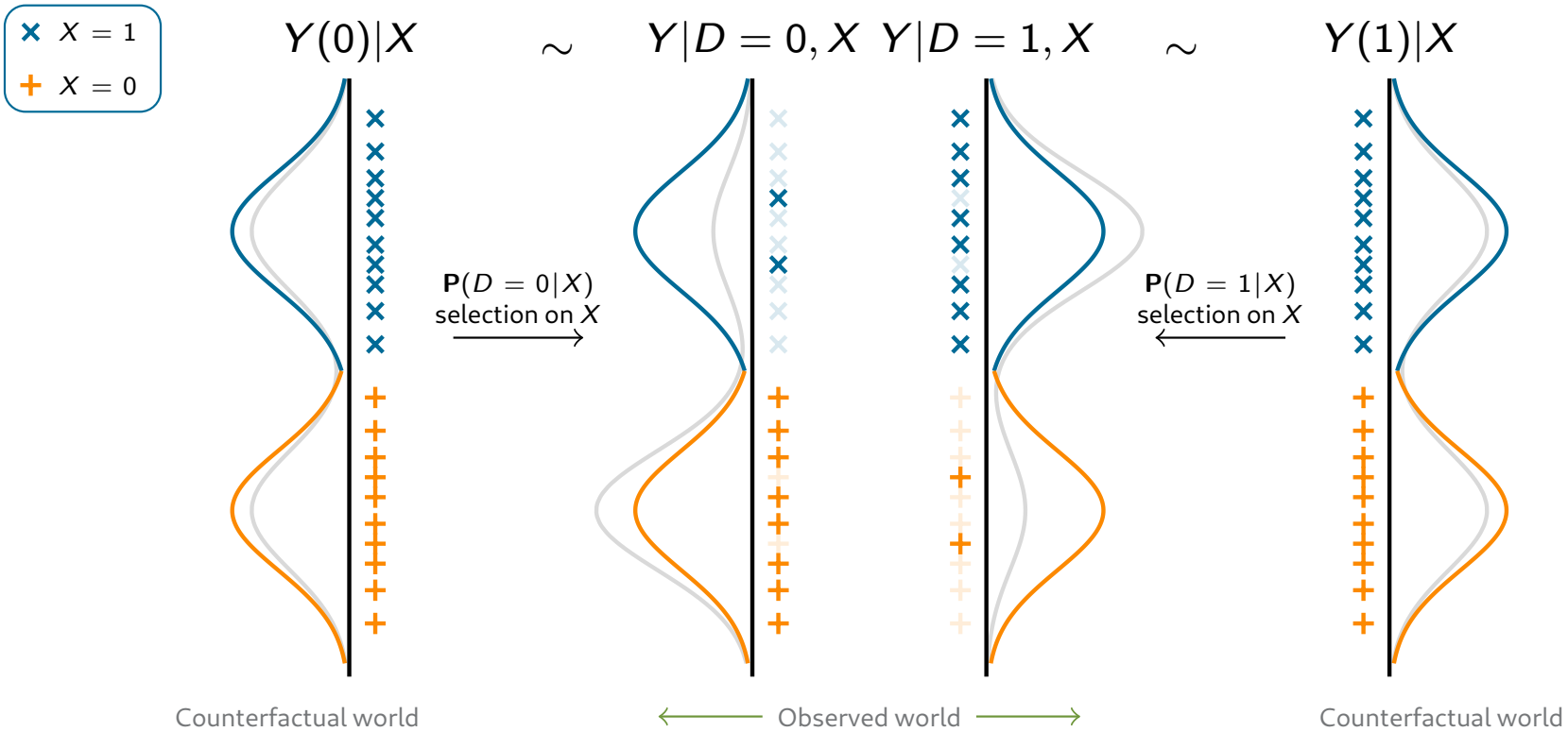


Within $X = x$, selection is as good as random: untreated $X = 1$ units look like all $X = 1$ units

$\Longrightarrow$ $Y|D = 0, X$ has the same distribution as $Y(0)|X$

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

Conditioning on X removes selection bias

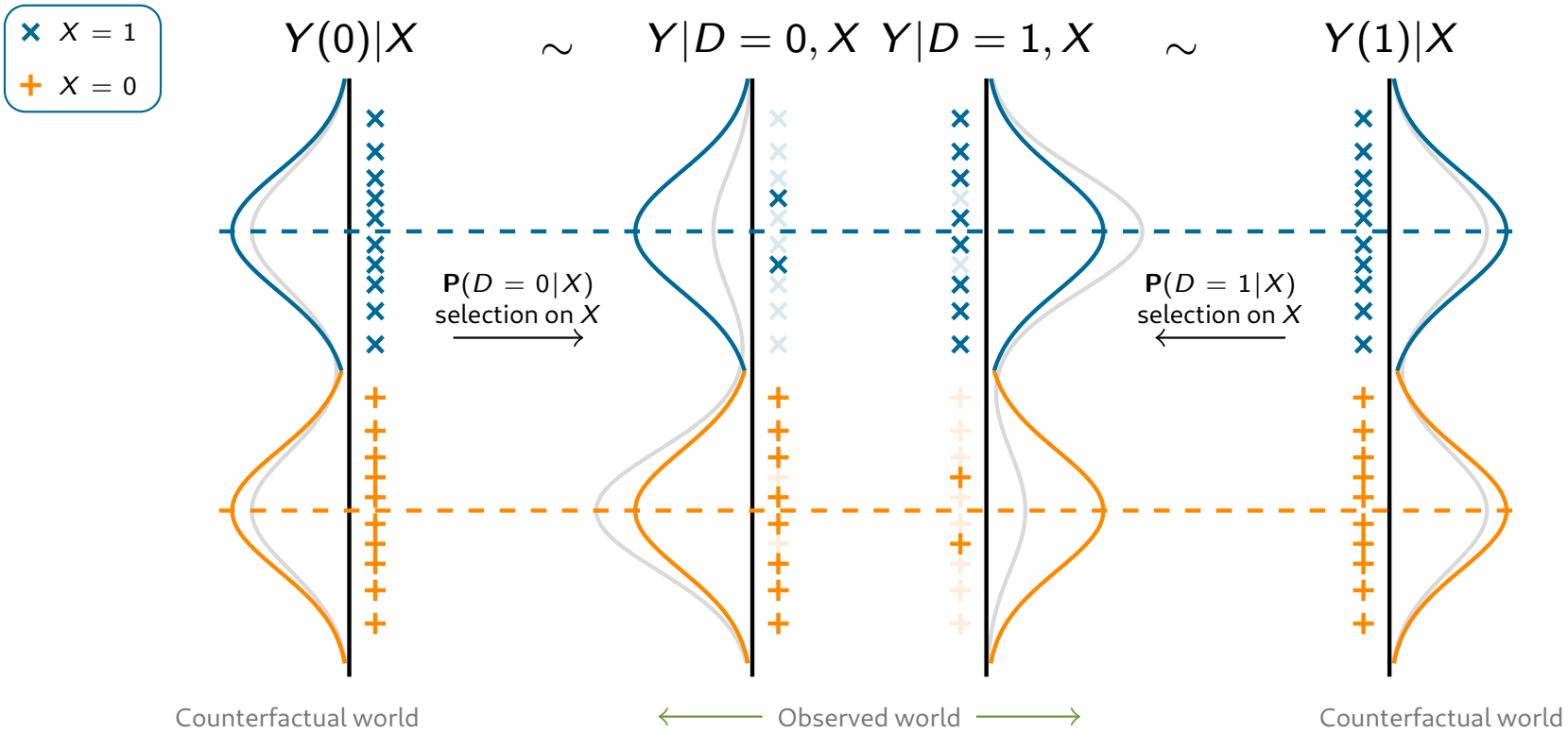


Same for the treated: $Y|D = 1, X$ has the same distribution as $Y(1)|X$

- Only the *mix* of X differs between treated and untreated, not outcomes within X

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

Conditioning on X removes selection bias



Within each $X = x$, observed means equal potential outcome means (dashed lines):

$$\begin{aligned}\mathbf{E}[Y|D = 1, X = x] - \mathbf{E}[Y|D = 0, X = x] &= \mathbf{E}[Y(1)|X = x] - \mathbf{E}[Y(0)|X = x] \\ &= \delta(x) = 0\end{aligned}$$

- Average $\delta(x)$ over the distribution of X to recover δ

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

Identification by conditioning

Ignorability says variation in D is **as good as random** conditional on X

- Among units with the same covariates ($X = x$), treatment variation $D|X = x$ is produced *as if* by an RCT
- "Ignorable": distribution of $Y(d)$ depends only on X, not on D
- Also called **conditional exogeneity** or the **conditional independence assumption** (CIA) in econometrics

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- Average these contrasts over the distribution of X to get a summary measure of the causal effect

For this to work, we need **both** treated and control units at every value of X

- The **propensity score** $p(X) := \mathbf{P}(D = 1|X)$ plays a key role

Source: Chernozhukov et al. (2025), Section 5.2

Overlap assumption

Overlap (full support): the propensity score is non-degenerate

$$p(X) := \mathbf{P}(D = 1|X), \qquad \mathbf{P}\big(0 < p(X) < 1\big) = 1$$

- Requires variation in D at each value x in the support of X
- Conditional analogue of **positivity** under random assignment ($0 < \mathbf{P}(D = 1) < 1$)

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Without overlap: at some values x there are only treated (or only control) units

- Cannot construct a contrast between treated and control at those x

$\Longrightarrow$ Cannot learn the conditional average treatment effect at those values of X

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Example: if every heavy drinker smokes marijuana, then $p(\text{heavy drinker}) = 1$

- No non-smoking heavy drinkers to compare against

Source: Chernozhukov et al. (2025), Assumption 5.2.2 and Remark 5.2.1

Identification theorem

Assumptions:

1. **Consistency:** $Y = Y(D)$
2. **Conditional ignorability:** $D \perp Y(d)|X$ for each d
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Theorem (conditioning on X removes selection bias): Under consistency, conditional ignorability, and overlap,

$$\mathbf{E}[Y|D = d, X] = \mathbf{E}[Y(d)|D = d, X] = \mathbf{E}[Y(d)|X]$$

- The CEF of observed Y given $D = d$ and X recovers the CEF of the potential outcome $Y(d)$ given X

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Why?

- Overlap makes it possible to condition on $\{D = 0, X\}$ and $\{D = 1, X\}$ at every value in the support of X
- First equality: consistency
- Second equality: ignorability

Source: Chernozhukov et al. (2025), Theorem 5.2.1

Predictive effects and treatment effects

Predictive effects (computable from observed data):

$$\text{Conditional avg. predictive effect (CAPE):} \quad \pi(X) = \mathbf{E}[Y|D=1,X] - \mathbf{E}[Y|D=0,X]$$
$$\text{Avg. predictive effect (APE):} \quad \pi = \mathbf{E}[\pi(X)]$$

Treatment effects (defined by potential outcomes):

$$\text{Conditional avg. treatment effect (CATE):} \quad \delta(X) = \mathbf{E}[Y(1)|X] - \mathbf{E}[Y(0)|X]$$
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By the theorem, the conditional average predictive effect equals the conditional average treatment effect:

$$\pi(X) = \delta(X)$$

Averaging over X, the APE and ATE also agree:

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Averaging over X, the APE and ATE also agree:

$$\delta = \mathbf{E}[\delta(X)] = \mathbf{E}[\pi(X)] = \pi$$

- $\pi(X)$ is computable from data $\implies$ we can learn $\delta(X)$ and δ
- Note: $\pi = \mathbf{E}[\pi(X)]$ generally differs from the naïve comparison $\mathbf{E}[Y|D=1] - \mathbf{E}[Y|D=0]$

Source: Chernozhukov et al. (2025), Section 5.2

When treatment takes on more than two values

Expanding CIA assumption to relations that can take on more than two values, such as when $s_i \in \{0, 1, \ldots, 16\}$ equals years of schooling:

$$Y_i(s) = f_i(s)$$

Individual specific function:

- Potential earnings that person i would receive after obtaining s years of education
- What would i earn for any value of schooling

If s could only take on two values, $s = 12$ and $s = 16$, then we're considering the value of college vs. no college:

$$Y_i(0) = f_i(12), \qquad Y_i(1) = f_i(16)$$

CIA with non-binary treatment

CIA in this more general set-up:

$$Y_i(s) \perp s_i | X_i, \quad \forall s$$

- In randomized experiments (RCTs), s_i is randomly assigned based on covariates X_i
- In observational studies, the CIA means s_i is **as good as randomly assigned** conditional on X_i

Under conditional independence assumption, comparisons of average earnings across schooling levels have a causal interpretation:

$$\mathbf{E}[Y_i|X_i, s_i = s] - \mathbf{E}[Y_i|X_i, s_i = s-1] = \mathbf{E}[f_i(s) - f_i(s-1)|X_i]$$

For example, average causal effect of high school graduation:

$$\begin{aligned}\mathbf{E}[Y_i|X_i, s_i = 12] - \mathbf{E}[Y_i|X_i, s_i = 11] =& \mathbf{E}[f_i(12)|X_i, s_i = 12] - \mathbf{E}[f_i(11)|X_i, s_i = 11] \\ =& \mathbf{E}[f_i(12) - f_i(11)|X_i, s_i = 12]\end{aligned}$$

- Here, selection bias would come from the differences in potential earnings of high school graduates and non-graduates
- Under CIA, high school graduation is independent of potential earnings conditional on X_i, so selection bias disappears

CIA and regression

Regression is an empirical strategy that turns CIA into causal effects:

- Assume $f_i(s)$ is linear in s, and the same for everyone, except for an additive term η_i:

$$f_i(s) = \alpha + \rho s + \eta_i$$

- **Linear constant effects model**: equation is linear, and the functional relationship is the same for everyone
- Only individual-specific part of $f_i(s)$ is mean-zero η_i, which captures unobserved factors that determine potential earnings

Observed value will be:

$$Y_i = \alpha + \rho s_i + \eta_i$$

Note that s_i is correlated with η_i: $\mathbf{E}[\eta_i | s_i] \neq 0$. If we ran this regression, we'd expect selection bias to impact our results.

CIA and regression (cont.)

Assume CIA holds, and η_i has the form:

$$\eta_i = X_i'\gamma + v_i$$

where:

- γ is a vector of population regression coefficients that satisfies $\mathbf{E}[\eta_i|X_i] = X_i'\gamma$
- v_i and X_i are uncorrelated by construction

Then, because of the CIA:

$$\begin{aligned}\mathbf{E}[f_i(s)|X_i, s_i] &= \mathbf{E}[f_i(s)|X_i] \\ &= \alpha + \rho s + \mathbf{E}[\eta_i|X_i] \\ &= \alpha + \rho s + X_i'\gamma\end{aligned}$$

The residual in the linear causal model v_i is uncorrelated with s_i and X_i, and ρ is the causal effect of interest:

$$Y_i = \alpha + \rho s_i + X_i'\gamma + v_i$$

Regression takeaways

We just discussed how CIA + linear causal model yields a causal interpretation of CEF

- Linear causal model: $Y_i = \alpha + \rho s_i + \eta_i$
- If CIA holds given some variables X_i: $Y_i(s) \perp s_i | X_i \; \forall s$
- Our regression function has a causal interpretation: $Y_i = \alpha + \rho s_i + X_i' \gamma + v_i$
  - ρ is the causal effect of s_i on Y_i
- CEF given by:

$$\mathbf{E}[Y_i | X_i, s_i] = \alpha + \rho s_i + X_i' \gamma$$

Things to note:

- CIA is the same as unconfoundedness is the same as selection on observables
- In regression, CIA is the same as exogeneity